Minor- III

School of Mathematics and Statistics
University of Hyderabad

October 27, 2026

Duration: 60 minutes
Maximum Score: 20 points

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Course code: ST 402

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

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1. Let t_1, t_2 and t_3 be any distinct real numbers.

   Define
   $$L_i \ : \ \mathcal{P}_2(\mathbb{R}) \ \longrightarrow \ \mathbb{R}$$
   such that
   $$L_i(p) \ := \ p(t_i), \quad \forall \ i = 1, \ 2, \ 3.$$

   (a) Show that $\{L_1, \ L_2, \ L_3\}$ is basis for $(\mathcal{P}_2(\mathbb{R}))^*$. [4]

   (b) Find the basis of $\mathcal{P}_2(\mathbb{R})$ of which $\{L_1, \ L_2, \ L_3\}$ is dual basis. [4]

2. Let W be proper subspace of finite - dimensional vector space V.

   Show that there exists a nonzero linear functional f on V such that $f \equiv 0$ on W. [4]

3. Let V and W be finite - dimensional vector spaces. Let $T \ : \ V \ \longrightarrow \ W$ be linear map.

   Show that
   $$\text{rank } (T) \ = \ \text{rank } (T^t).$$
   [4]

4. Consider the following linear operator on $\mathcal{P}_n(\mathbb{R})$:
   $$D(a_0 \ + \ a_1 x + \cdots + \ a_n x^n) \ = \ a_1 \ + \ 2a_2 x \ + \cdots + \ n a_n x^{n-1}.$$
   Find a basis for the null space of D^t. [4]

All the best !